

ASSIGNMENT – 1

Introduction to Sensors and Actuators (UEC 3942)

AIM

To design sensor-based systems for twenty real-world application scenarios, selecting appropriate sensors, signal conditioning, processing, and display/actuation elements for each. The designs cover temperature, optical, pressure, strain, and piezoelectric sensing principles.

Q1. Automatic Temperature Monitoring and Control for a Cold Storage Warehouse (RTD/Thermistor)

A PT100 RTD senses the warehouse temperature and a PID controller on a microcontroller holds the room between $+2^{\circ}\text{C}$ and -18°C by switching the refrigeration compressor through a solid-state relay.

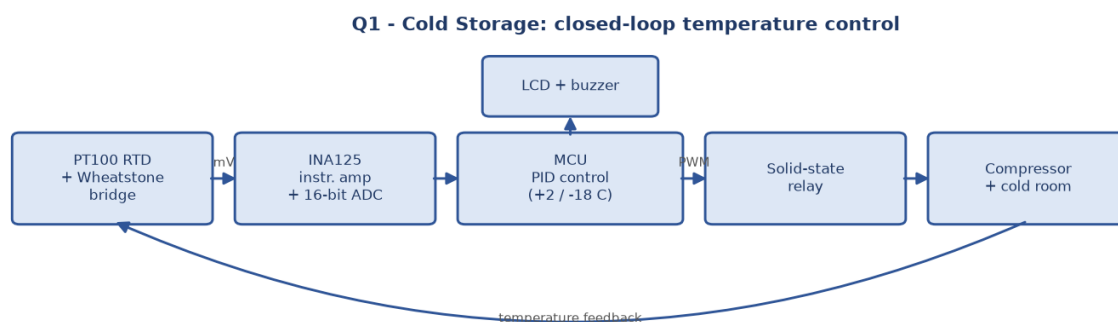


Fig 1. Closed-loop temperature control for the cold storage warehouse



PT100 RTD probe and bare sensing elements

- Sensing: PT100 RTD ($\pm 0.1^\circ\text{C}$, nearly linear) in a three-wire Wheatstone bridge to cancel lead resistance.
- Conditioning: INA125 instrumentation amplifier scales the bridge output to the ADC range.
- Control: microcontroller (ATmega328P/ESP32) runs a PID loop with 2°C hysteresis to stop compressor short-cycling.
- Output: SSR drives the compressor; LCD shows live temperature; buzzer alarms on excursions.
- Extra: door sensor raises an alarm if the door stays open over 30 seconds.

Q2. Fiber-Optic Security System for Detecting Unauthorized Entry

An infrared beam carried on optical fiber is stretched across the entry path; breaking the beam instantly triggers the alarm chain.

- Sensing: through-beam pair – 850 nm IR LED feeding the transmitter fiber, photodiode on the receiver fiber.
- Interference rejection: LED modulated at 38 kHz so the receiver only accepts the modulated signal (sunlight/ambient rejected).
- Detection: LM358 comparator flags the beam break and fires a microcontroller interrupt.
- Response: siren + CCTV trigger + magnetic door lock; latching relay holds the alarm until a key-switch reset.
- Why fiber: immune to EMI, no electrical conductor at the sensing point, works around metal doors and machinery.

Q3. Solar Energy Monitoring System Using a Photovoltaic Sensor

A calibrated PV cell measures solar irradiance while an MPPT converter keeps the panel at its maximum power point; everything is logged to the cloud.

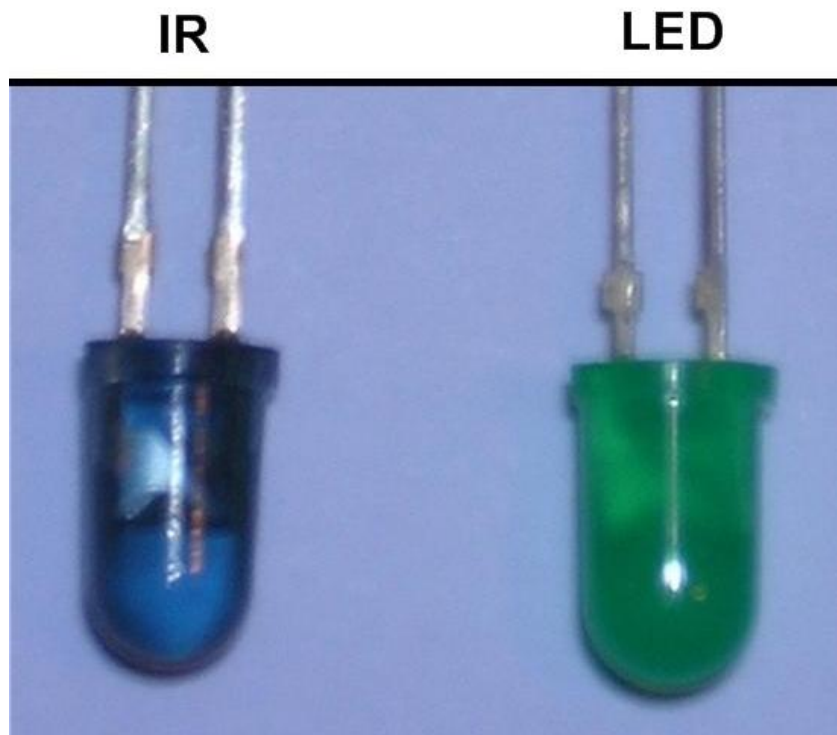
- Sensing: BPW34 photodiode in photovoltaic mode – short-circuit current is proportional to irradiance ($0\text{--}1000\text{ W/m}^2$).
- Conditioning: OP07 transimpedance amplifier (10 V/A) followed by a 1 s RC low-pass filter against cloud flicker.
- Conversion: 16-bit ADC (ADS1118) with calibration constant converts to W/m^2 .
- Power tracking: ESP32 runs perturb-and-observe MPPT, adjusting a DC-DC buck converter duty cycle.
- Logging: irradiance, voltage, current, power published over MQTT; local LCD for field checks.

Q4. Optical Fire Detection System Using a Photodiode/Phototransistor for Industrial Safety

A dual-wavelength optical detector senses both smoldering fires (IR light scattered by smoke) and open flames (visible flicker), latching an alarm and reporting over wireless.

Q4 - Optical Fire Detector: scatter chamber principle

Fig 2. Optical fire detector using 90-degree light scatter chamber

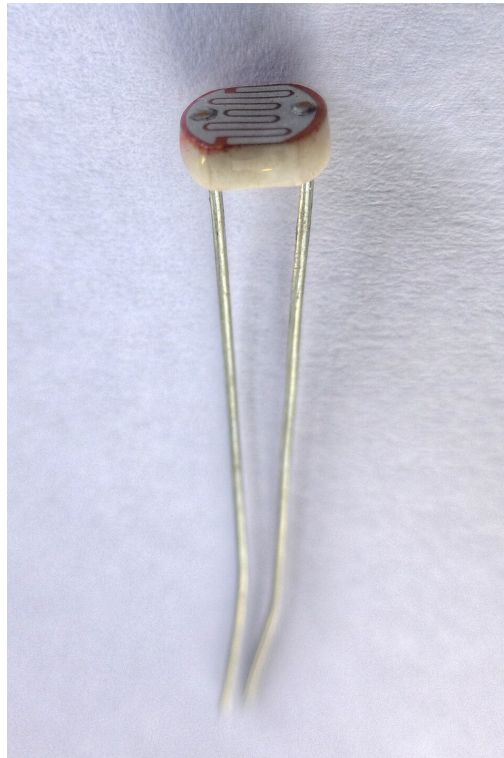


Infrared emitting LEDs used as the chamber source

- Smoke channel: 880 nm IR LED and phototransistor at 90° in a dark chamber – smoke particles scatter light onto the detector.
- Flame channel: 660 nm band photodiode detects the characteristic 1–5 Hz flicker of real flames (steady lamps/sun rejected).
- Threshold: alarm at ~4.5%/ft obscuration (scatter coefficient > 0.3 m⁻¹) with hysteresis to stop chattering.
- Alerting: 85 dB piezo sounder + beacon; state broadcast over LoRa to the plant BMS.
- Self-test: button injects a known scatter signal to verify the whole chain.

Q5. Automatic Brightness Control Circuit for an LCD Display Using a Photo Sensor

An ambient light sensor feeds a PI control loop that continuously trims the backlight PWM so panel luminance stays constant as room light changes.



LDR (photoresistor) used as the low-cost ambient light sensor

- Sensing: BPW34 photodiode or LDR voltage divider beside the display; ADC reading converted to lux.
- Control: PI loop against a comfort setpoint ($\sim 200 \text{ cd/m}^2$); $\sim 1 \text{ s}$ settling so changes never distract.
- Drive: backlight LED driver PWM at $> 200 \text{ Hz}$ (no visible flicker), duty cycle 0–100%.
- Automotive variant: two sensors – road-facing (glare) and sky-facing (ambient) – blended for stability.
- Result: readable in sunlight, dim at night, and lower power draw.

Q6. Barcode Scanning System Using Optical Sensors

Red illumination and a line sensor capture the bar pattern, and a decoder converts the reflected waveform into the product code.

- Illumination: 660 nm red LED – dark bars absorb, white background reflects strongly.
- Capture: CCD line sensor, or a laser spot swept by a mirror galvanometer onto one photodiode.
- Decoding: comparator thresholds the waveform; DSP measures bar/space widths and matches symbol geometry.
- Output: UART/USB-HID string to the POS system.
- Specs: 45° field of view, 50–200 mm working distance; camera alternative (OV7670) also reads QR codes.

Q7. Smart Traffic Signal Controller Adjusting Brightness with Ambient Light

Photocells on each signal head dim the LEDs at night and brighten them in daylight, while the same controller adapts signal timing to traffic.

- Brightness: LDR on each head; full current in daylight, 30–50% at night to cut glare and power.
- Stability: switch thresholds at 50 lux down / 80 lux up (hysteresis against dusk oscillation).
- Timing: day mode gives longer green to heavier approaches via vehicle loops; night mode shortens cycles.
- Preemption: 38 kHz IR receiver lets ambulances/fire trucks force a green corridor.
- Safety: min green 10 s, max 60 s; watchdog forces all-red flash if sensor feedback is lost.

Q8. Temperature and Pressure Monitoring System for a Steam Pipeline (Thermocouple + Pressure Sensor)

A Type K thermocouple and a 4–20 mA pressure transmitter watch the steam line, with alarms and trends on SCADA.



Type K thermocouple probe used for steam line measurement

- Temperature: Type K thermocouple (–200°C to +1350°C) in an Inconel thermowell; MAX31855 does cold-junction compensation.
- Pressure: 4–20 mA transmitter (0–100 bar); 250 Ω sense resistor gives 1–5 V, and <4 mA signals a broken wire.
- Acquisition: both channels sampled at 1 Hz by a DAQ module.
- Display: SCADA panel with live values and 24 h trend curves.
- Alarms: SMS/email above 400°C or 80 bar, all events timestamp-logged.

Q9. Pulse Monitoring System Using Optical Sensors (Photodiode/Phototransistor)

Light through a fingertip rises and falls with each heartbeat; filtering that tiny signal out of the large DC baseline recovers the pulse waveform and BPM.

Q9 - Pulse Monitor: photoplethysmography chain

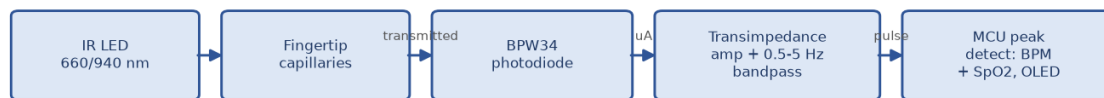


Fig 3. Photoplethysmography signal chain of the pulse monitor

- Principle: photoplethysmography – blood volume modulates light absorption (Beer-Lambert law).
- Sensing: IR LED one side of the finger, BPW34 photodiode on the other.
- Conditioning: transimpedance amplifier, then 0.5–5 Hz bandpass removing DC drift, respiration and mains hum.
- Processing: peak detection on the filtered pulse; BPM = 60 / RR-interval, averaged over beats.
- Display: 128×64 OLED shows waveform + BPM; adding 660 nm + 940 nm LEDs enables SpO.

Q10. Optical Smoke Detector for Residential Buildings (Photodiodes and Infrared LEDs)

A photoelectric chamber detects smoke by the infrared light it scatters toward a photodiode, sounding a loud alarm while sipping battery current.

- Chamber: 850 nm IR LED and photodiode at 90°; clean air means no light reaches the detector.
- Thresholds: alarm at 4.5%/ft obscuration, reset at 3.5%/ft (hysteresis); sensitivity ~1.5%/ft.
- Power: pulsed LED (burst every 10 s) gives 5-year life on 3×AA cells.
- Features: 15-minute hush button; 433 MHz wireless interconnect sounds every unit in the house.
- Testing: button injects a known scatter signal; verify periodically with aerosol smoke.

Q11. Automatic Museum Lighting System to Protect Artifacts from Light Exposure

Lux and motion sensors let the lighting follow visitors and daylight, holding illumination on artifacts inside conservation limits.

- Sensing: TCS34725 lux/color sensor at exhibit level; PIR (HB100) for visitor presence.
- Conservation: total light held at 150–300 lux; daylight harvesting dims LEDs when sun exceeds 300 lux.

- Occupancy: lights fade to 10% after 5 min empty; 5 s ramp-up when someone approaches.
- Quality: fixtures with CRI > 90 plus UV/IR filtering (the main pigment damagers).
- Control/energy: DMX512 zoning per case; 10% security lighting after hours; 60–80% energy saving.

Q12. Portable Digital Body Temperature Monitoring Device

A non-contact infrared thermopile reads forehead temperature in a second, with compensation and averaging for clinical accuracy.



Infrared thermometer of the type built around a thermopile sensor

- Sensing: MLX90614 IR thermopile, -40°C to $+300^{\circ}\text{C}$, $\pm 0.2^{\circ}\text{C}$ in the body range, internal 16-bit ADC.
- Compensation: ambient (die) temperature correction curve for forehead-to-core offset.
- Processing: 5 Hz sampling, 5-reading moving average with outlier rejection.
- UI: 1.3" TFT at 0.1° resolution; green/orange/red result; beep above 38°C fever threshold.
- Power: 2×AAA for 1000+ readings; factory calibration with $\pm 0.5^{\circ}\text{C}$ user offset.

Q13. Industrial Monitoring System Measuring Stress, Pressure, and Temperature Simultaneously

Strain gauges, a pressure transmitter and an RTD feed one time-aligned acquisition chain that streams engineering-unit data to the cloud with edge alarms.

- Stress: 350 foil strain gauges in Wheatstone bridge (full-bridge cancels bending errors), INA125 amplifier.
- Pressure: 4–20 mA loop (0–100 bar) into 250 sense resistor.

- Temperature: PT100 RTD in its own bridge.
- Acquisition: 74HC4051 multiplexer into a 16-bit ADS1117 (860 S/s) keeps channels time-aligned.
- Edge/cloud: Raspberry Pi 4 converts to engineering units, alarms at strain $>2000 \mu$, publishes MQTT, buffers locally on outage; Zigbee mesh backhaul.

Q14. Automatic Vehicle Headlight Control System Using LDR and Phototransistor

Two light sensors decide when the headlights belong on, and a servo auto-levels the beam so it never dazzles oncoming drivers.

- Sensing: sky-facing LDR (ambient) + road-facing phototransistor (tunnels, oncoming glare).
- Logic: on below 200 lux, off above 100 lux (hysteresis) with 5 s delay against overpass flicker.
- Drivers: halogen via relay; HID via DC-DC boost + igniter; LED via 0–100% PWM dimming.
- Auto-leveling: ADXL345 pitch sensing drives beam-height servos when the rear is loaded.
- Override: manual switch always wins for driver confidence.

Q15. Structural Health Monitoring System for a Bridge Using Resistive Strain Gauges

Strain gauges at the bridge's critical sections stream load data continuously, watching both overload and stiffness changes that reveal damage.

- Placement: midspan bottom flanges (max positive bending), over piers (negative bending), main cables/trusses.
- Compensation: temperature-compensated half/full bridges; co-located RTD subtracts thermal strain ($= LT$).
- Acquisition: instrumentation amps + 24-bit ADCs at 1 Hz per node.
- Alarms: quasi-static strain $>2000 \mu$ (0.2%), strain-rate spikes (impact), FFT natural-frequency drift (stiffness loss).
- Network: 802.15.4 Zigbee mesh to a cellular gateway; low duty cycle gives 10-year battery life.

Q16. Electronic Weighing Scale Using Resistive Strain Gauges and a Wheatstone Bridge

A four-gauge full-bridge load cell turns weight into a tiny differential voltage that amplification and a 24-bit ADC resolve to 1 g.

Q16 - Electronic Weighing Scale: full-bridge load cell

Fig 4. Full-bridge load cell measurement chain of the weighing scale

- Load cell: aluminum bending beam with 4× 350 gauges (2 tension, 2 compression) – 4× sensitivity, self-compensating.
- Signal: ~10 mV full scale (2 mV/V at 5 V excitation); INA125 at gain 100 1 V.
- Conversion: ADS1256 24-bit sigma-delta at 2000 S/s, ~200 ms averaging for a stable reading.
- Calibration: EEPROM zero (tare) + span with certified 5 kg weight; ±0.1% FS accuracy.
- UI: LED display, auto-tare button, RS232 to POS; mechanical overload stop at 150%.

Q17. Piezoelectric Sensor-Based Vibration Monitoring for Engine Knock Detection

A piezoelectric sensor on the engine block hears the characteristic 5–10 kHz ringing of knock, and the ECU retards timing until it stops.

- Sensing: PZT piezo element converts block vibration to charge ($Q = d \cdot F$).
- Conditioning: charge amplifier (~10 V/C effective) for the high-impedance source; bandpass 5–10 kHz isolates knock resonance.
- Gating: signal windowed per cylinder from the crank-angle sensor; RMS computed in-window.
- Control: knock declared above ~1.5 V RMS (noise-adaptive); ignition retarded 2°–8° CA, then advanced back gradually.
- Logging: Arduino Mega-class MCU with fast ADC; per-cylinder knock intensity to SD card and LCD.

Q18. Wireless Tire Pressure Monitoring System (TPMS) Using Pressure Sensors

A battery module inside each wheel measures pressure and temperature and radioes it to the dashboard, warning the driver before a tire becomes dangerous.

Q18 - TPMS: wireless tire pressure module

Fig 5. TPMS wireless transmission link from wheel module to dashboard



TPMS sensor strapped inside the wheel on the rim

- Sensing: MEMS absolute pressure sensor (0–8 bar) + thermistor for internal air temperature.
- Power: accelerometer wakes the module only when the wheel rolls; CR1620 cell lasts 2+ years.
- Link: 433 MHz ASK/OOK frame with unique 32-bit ID, pressure, temperature, battery status; ~30 ft range.
- Matching: ECU learns wheel positions by deflate/inflate relearn or stationary magnet tool (20 min).
- Compliance: FMVSS 138 – warn at 25% below placard pressure; fast-leak (>1 psi/s) triggers an immediate red warning.

Q19. Real-Time Pressure Monitoring System for an Industrial Boiler

A HART pressure transmitter with redundant sampling watches drum pressure, with a hard-wired safety interlock as the last line of defense.

- Sensing: 4–20 mA transmitter, 0–250 bar, $\pm 0.1\%$ FS; HART digital overlay carries diagnostics and trimmed values.
- Redundancy: 250 sense resistor 1–5 V sampled on two ADC channels with voting logic.
- Local UI: LCD with bar graph at the boiler, independent of the control network.
- Alarms: SMS to plant manager at 90% setpoint; at 110% a hard-wired interlock trips the burner and feedwater (no software in the path).
- Records: SCADA historian (InfluxDB) every 10 s; independent bourdon-tube gauge as regulatory backup.

Q20. Design of an Automatic Temperature Monitoring and Control System for a Cold Storage Warehouse Using an RTD or Thermistor (Variant B)

This variant uses an NTC thermistor instead of an RTD, trading a little accuracy for much lower cost – suitable for a small commercial cold room.

- Sensing: 10 k NTC thermistor in a voltage divider; Steinhart-Hart equation linearizes the strongly nonlinear R-T curve in firmware.
- Conditioning: divider buffered by an op-amp voltage follower into the ADC (no bridge needed).
- Control: same PID + hysteresis structure on the compressor SSR; setpoints +2°C / -18°C.
- Extras: capacitive hygrometer (HSM20G) for humidity, door-open alarm >30 s, data-logged excursions.
- Trade-off: thermistor $\pm 0.5^{\circ}\text{C}$ over 0–40°C and lower cost vs RTD $\pm 0.1^{\circ}\text{C}$ and better long-term stability.

LEARNING OUTCOMES

- Selected sensing elements (RTD, thermocouple, photodiode, LDR, strain gauge, piezoelectric, MEMS) by range, accuracy and environment.
- Designed complete signal chains: bridges, instrumentation/charge/transimpedance amplifiers, filtering, ADC choice.
- Applied modulation, hysteresis and thresholding to reject interference and stop control chattering.
- Integrated microcontroller processing: PID loops, FFT analysis, peak detection, MPPT.
- Specified end-to-end systems from sensor to display, alarm and cloud/SCADA reporting.

REFERENCES

- D. Patranabis, Sensors and Transducers, PHI Learning.
- C. R. Webb, Sensors and Actuators: Engineering System Instrumentation, Elsevier.
- J. Fraden, Handbook of Modern Sensors, Springer.
- UEC 3942 Introduction to Sensors and Actuators, SSN College of Engineering, Regulations 2024.